

OE-EC804B — Microwave Integrated Circuits

Semester VIII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Planar Lines-II (Slotline, Finlines & CPW)

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1 Slot Line Analysis and Field Distribution

A **Slotline** consists of a narrow slot of width W etched in a conductive metal plane coated on one side of a dielectric substrate (ϵ_r of thickness h), leaving the reverse side unmetallized.

Slotline Structure & Propagation Mode

Slotline is a non-TEM planar transmission line supporting a **TE-like mode**. Electric field lines extend horizontally across the slot gap (E_y), while magnetic field lines form closed loops in the longitudinal plane ($H_z \neq 0, H_x \neq 0$). Because $H_z \neq 0$, slotline exhibits elliptical magnetic field polarization, making it ideally suited for integrating non-reciprocal ferrite components (such as isolators and circulators).

1.1 Transverse Resonance Method (TRM) Analysis

Because slotline fields are non-TEM, electro-static Laplace equations fail to predict dispersion. The **Transverse Resonance Method (TRM)** models the transverse cross-section as an equivalent transmission line resonant system in the transverse y -direction.

Transverse Resonance Condition

Consider a transverse cut plane at $y = 0$ across the slot interface. At resonance, the sum of input admittances looking into the upper air region ($\bar{Y}_{up}$) and lower substrate region ($\bar{Y}_{down}$) must sum to zero:

$$\bar{Y}_{up}(y = 0) + \bar{Y}_{down}(y = 0) = 0 \quad (1)$$

Enforcing $\text{Im}\{Y_{total}(\beta)\} = 0$ yields the guided wavenumber $\beta = 2\pi/\lambda_g$. Characteristic impedance Z_0 is evaluated from root-mean-square slot voltage V :

$$Z_0 = \frac{V^2}{2P_{avg}} \quad \text{where } V = \int_{-W/2}^{W/2} E_y(x) dx \quad (2)$$

1.2 Exhaustive Comparison: Slotline vs. Microstrip Line

Parameter	Microstrip Line	Slotline
Dominant Mode	Quasi-TEM mode ($E_z, H_z \approx 0$)	Non-TEM / TE-like mode ($H_z \neq 0$)
Field Distribution	Electric fields confined under central strip	Electric fields concentrated across slot gap
Shunt Mounting	Requires via holes drilled through substrate to ground	Direct surface mounting of active dies across slot without via holes
Series Mounting	Easy in-line series trace breaking	Difficult series connection
Impedance Range	20 Ω to 120 Ω (Low impedance lines)	50 Ω to 300 Ω (High impedance lines)
Magnetic Polarization	Linear transverse magnetic fields	Elliptically polarized H -field (Ideal for Ferrite devices)
Radiation Loss	Moderate at microwave frequencies	Slightly higher due to wide open top slot

2 Coplanar Waveguide (CPW) and Fin Lines

2.1 Coplanar Waveguide (CPW)

A **Coplanar Waveguide (CPW)** consists of a central signal conductor of width W flanked by two semi-infinite ground planes on the **same top surface** of a dielectric substrate (ϵ_r), separated by narrow gaps S .

Coplanar Waveguide (CPW) Advantages

CPW supports a Quasi-TEM mode. Because both signal conductor and ground planes reside on the same face, both series and shunt components can be mounted **without drilling via holes**, significantly reducing

MMIC manufacturing cost and parasitic via-inductance ($L_{via} \approx 0.1 \text{ nH}$).

2.2 CPW Conformal Mapping & Impedance Formulas

Using conformal mapping, CPW effective dielectric constant $\varepsilon_{eff,CPW}$ and characteristic impedance $Z_{0,CPW}$ are derived in terms of complete elliptic integrals of the first kind $K(k)$:

CPW Closed-Form Impedance Expressions

$$\varepsilon_{eff,CPW} = \frac{\varepsilon_r + 1}{2} \quad ; \quad Z_{0,CPW} = \frac{30\pi}{\sqrt{\varepsilon_{eff,CPW}}} \frac{K'(k)}{K(k)} \quad (3)$$

where modulus $k = \frac{W}{W+2S}$ and $K'(k) = K(k')$ with $k' = \sqrt{1-k^2}$.

2.3 Fin Lines (*E*-Plane Waveguide Printed Lines)

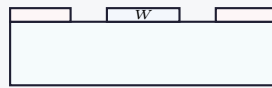
A **Finline** consists of a slotline printed on a thin dielectric substrate mounted along the central longitudinal *E*-plane of a rectangular metallic waveguide housing.

1. **Unilateral Finline:** Slot printed on one face of the substrate only.
2. **Bilateral Finline:** Identical slots printed symmetrically on both substrate faces.
3. **Insulated Finline:** Substrate insulated from metallic waveguide top/bottom walls.

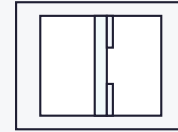
Cross-Sections of Slotline, CPW, and Unilateral Finline



(a) Slotline



(b) Coplanar Waveguide (CPW)



(c) Finline

Worked Numerical: Coplanar Waveguide Impedance Calculation

Problem: A Coplanar Waveguide (CPW) is fabricated on a GaAs substrate ($\varepsilon_r = 12.9$). Calculate:

- (a) The effective dielectric constant $\varepsilon_{eff,CPW}$.
- (b) The characteristic impedance Z_0 if ratio $K'(k)/K(k) = 1.0$.

Solution:

- (a) Effective dielectric constant for symmetrical CPW:

$$\varepsilon_{eff,CPW} = \frac{\varepsilon_r + 1}{2} = \frac{12.9 + 1}{2} = 6.95 \quad (4)$$

- (b) Characteristic impedance calculation:

$$Z_{0,CPW} = \frac{30\pi}{\sqrt{6.95}} \times (1.0) = \frac{94.2477}{2.63628} = 35.75 \, \Omega \quad (5)$$

Quick Revision — Module 3

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Key Concepts and Metrics at a Glance

Slotline Mode	Non-TEM / TE-like mode with longitudinal magnetic field $H_z \neq 0$.
Transverse Resonance Method	Evaluates slotline & finline parameters via admittance resonance $\sum Y_{transverse} = 0$.
CPW Effective Dielectric	$\epsilon_{eff,CPW} = \frac{\epsilon_r + 1}{2}$ (Equal electric energy split in air and substrate).
CPW Impedance	$Z_{0,CPW} = \frac{30\pi}{\sqrt{\epsilon_{eff}}} \frac{K'(k)}{K(k)}$ (Using complete elliptic integrals).
Via-Hole Elimination	CPW and Slotline allow direct surface shunt mounting of active devices without via holes.

One-Line Definitions

- **Slotline:** A planar line formed by a narrow slot in a conductive ground plane on a dielectric substrate.
- **Coplanar Waveguide (CPW):** A planar line with center conductor and dual ground planes on the same surface.
- **Finline:** An E -plane slotline enclosed inside a rectangular waveguide wall.
- **Transverse Resonance Method (TRM):** Analytical field method enforcing zero net input admittance across transverse cut planes.

Frequently Asked / Viva Questions

Q1. What is the dominant propagation mode in a slotline?

Slotline supports a non-TEM TE-like mode with a transverse electric field E_y and longitudinal magnetic field H_z .

Q2. Compare Slotline and Microstrip line.

Microstrip uses a Quasi-TEM mode with $20\ \Omega$ – $120\ \Omega$ impedance and requires via holes for shunt mounting. Slotline uses a TE-like mode with $50\ \Omega$ – $300\ \Omega$ impedance and allows direct surface shunt mounting.

Q3. Why does Coplanar Waveguide (CPW) eliminate the need for via holes?

Because both signal conductor and ground planes reside on the top surface of the substrate, allowing active devices to be connected directly in shunt.

Q4. Explain the Transverse Resonance Method (TRM).

TRM models the cross-section of a non-TEM transmission line as a transverse resonant line, finding β by setting total transverse input admittance $\sum Y(y) = 0$.

Q5. What is a Finline? Name its three main structural variations.

A Finline is an E -plane slotline enclosed inside a metallic rectangular waveguide. Variations: Unilateral, Bilateral, and Insulated Finlines.

Q6. Why is slotline suitable for mounting non-reciprocal ferrite isolators?

Because the slotline magnetic field exhibits circular/elliptical polarization in the substrate region, ideal for ferrite interaction.

Q7. How does the characteristic impedance of a CPW depend on gap width S ?

Frequently Asked / Viva Questions (contd.)

Increasing gap width S decreases total line capacitance, thereby increasing characteristic impedance Z_0 .

Q8. What is an air-bridge in CPW circuits?

A raised metal wire bridge connecting the dual ground planes of CPW to suppress unwanted odd-mode (slotline mode) propagation.

Q9. What are the advantages of Finlines at millimeter-wave frequencies (> 30 GHz)?

Combines the low loss and high power handling of rectangular waveguides with the planar circuit integration of slotlines.

Q10. How does conductor loss α_c in Finlines vary with slot width W ?

Conductor loss increases inversely with slot width ($1/W$) due to high current density concentration at fin edges.